

Pranav S. Kulkarni

Postdoctoral Researcher

POPULATION HEALTH AND REPRODUCTION,
SCHOOL OF VETERINARY MEDICINE,
UNIVERSITY OF CALIFORNIA DAVIS

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Research Interests

Epidemiology | One health | Veterinary Epidemiology | Farm animal health | Animal health Economics | Evidence based Veterinary Medicine | Biostatistics and Operations research

Education

Ph.D.	Jan 2024
Wageningen University & research, Utrecht University Thesis title: Dairy cattle replacement under changing agri-environmental policies ; ISBN: 978-94-6447-990-4 (Co-)Promoters: Prof.Dr. H. Hogeveen, Prof.Dr. M. Nielen, Dr. M.C.M Mourits, Dr. W. Steeneveld	
M.Sc.	Aug 2018
Wageningen University & research Major thesis: Genetic Epidemiology of Digital Dermatitis in Dutch dairy cattle Minor thesis: Simulation of density dependent and frequency dependent transmission of environmental pathogens Advisor: Dr. P. Bijma, Prof.Dr. M.C.M de Jong	
B.V.Sc & A.H.	Jul 2015
Maharashtra Animal and Fishery sciences University, Nagpur, India	

Research Experience

Postdoctoral researcher	Oct 2023-Present
(1) One Health Institute, UC-Davis (2) Dept. of Population Health and Reproduction, SVM, UC-Davis Ongoing projects [Wellcome Trust] "Atlas Arena" – (PI: Dr. P. Pandit) Description: Modeling the climate-driven zoonotic spillover risk of New World Arenaviruses. Combining human risk with changing species distribution models of reservoir rodent species. [USDA] "Modeling the transmission dynamics of HPAI in dairy cows" – (PI: Dr. P. Pandit) Description: Developing a nested agent-based model for mapping the transmission dynamics of Avian Influenza outbreak in dairy cow herds in California	
Doctoral researcher	Oct 2019-Jan 2024
(1)Wageningen University & Research, (2) Utrecht University PhD Project: "Optimization of replacement decisions in dairy cattle" Duties: Research resulting in publications, graduate and undergraduate thesis supervision/ assistance, teaching assistance	
M.Sc. Minor thesis	May 2018-Aug 2018
Quantitative Veterinary Epidemiology group, Wageningen University & Research (4 months: 24 ECTS) Title: "Simulation of density dependent and frequency dependent transmission of environmental pathogens" Supervisor: Prof. Dr. Mart de Jong	

M.Sc. Major thesis	Dec 2017-May 2018
Animal Breeding and genetics group, Wageningen University & Research (6 months: 36 ECTS) Title: “Genetic epidemiology of Digital dermatitis in Dutch dairy cattle” Supervisor: Dr. Piter Bijma	
Research Masters Cluster	Sep 2017-Nov 2017
Animal Sciences, Wageningen University & Research Training objectives: Writing grant proposals, formulating project objectives, pitching and brainstorming doctoral thesis ideas	
Undergraduate research assistant	2011-2015
Dept. of Animal Genetics and Breeding, KNP College of Veterinary Sciences, Maharashtra. Supervisor: Dr. Tejas Shende	

Publications

Pranav S. Kulkarni, Nuri Y. Flores-Perez, Andie H. Jian, Brian H. Bird, Christine K. Johnson, Marcela Uhart, Pranav S. Pandit (2026) [In Press] Climate-driven changes in zoonotic risk of arenaviral hemorrhagic fevers in South America., npj Viruses 4, 23 (2026). <https://doi.org/10.1038/s44298-026-00189-2>

Flores NY, Kulkarni P, Uhart M, Pandit PS (2025) Change Impact on Human-Rodent Interfaces: Modeling Junin Virus Reservoir Shifts. EcoHealth., <https://doi.org/10.1007/s10393-025-01723-z>

PS Kulkarni, R Haijema, H Hogeveen, W Steeneveld, MCM Mourits (2024) Economic impacts of constrained replacement heifer supply in dairy herds, Agric. Syst., 217 (103943), <https://doi.org/10.1016/j.agsy.2024.103943>

Kulkarni PS, Mourits M, Slob J, Veldhuis A, Nielen M, Hogeveen H, van Schaik G, Steeneveld W (2023) Dutch dairy farmers’ perspectives on culling reasons and strategies. Pre. Vet. Med. 218, <https://doi.org/10.1016/j.prevetmed.2023.105997>

Kulkarni PS, Mourits M, Nielen M and Steeneveld W (2023) Associations between dairy farm performance indicators and culling rates under policy-driven herd size constraints. Front. Vet. Sci. 10:1062891. <https://doi.org/10.3389/fvets.2023.1062891>

P Kulkarni, M Mourits, M Nielen, J van den Broek, W Steeneveld, (2021) Survival analysis of dairy cows in the Netherlands under altering agricultural policy, Pre. Vet. Med. 193, <https://doi.org/10.1016/j.prevetmed.2021.105398>

PS Kulkarni, F Biemans, MCM de Jong, P Bijma, (2021) On the origin of the genetic variation in infectious disease prevalence: Genetic analysis of disease status versus infections for Digital Dermatitis in Dutch dairy cattle, J. Ani. Bre. Gen. 138 (6), 629-642, <https://doi.org/10.1111/jbg.12635>

V Kadam, A Barate, T Shende, S Jadhav, P Kulkarni, P Pawar, M Rangnekar, H Birade (2020) Association of semen attributes with FSH – Beta gene polymorphism in Holstein-Frisian cross breed bulls from India, Haryana Vet, 59 (2), 231-235, <https://www.luvas.edu.in/haryana-veterinarian/download/harvet2020-dec/20.pdf>

TC Shende, PS Kulkarni, P Pawar, (2017) Genotyping of HF Crossbred cattle for Beta-Casein genes using PCR-RFLP, Indian Research Journal of Extension Education, 105-107, <https://shorturl.at/EG6Xz>

Preprints/ Working papers/ Conference proceedings

P Pandit, AH Jian, A Gupta, P Kulkarni, [pre-print] Ecological Cascades and Future Hantavirus Spillover Risk in a Changing Climate, bioRxiv 2026.07.13.738318; doi: <https://doi.org/10.64898/2026.07.13.738318>

S Dunakhe, A Sharma, A Das, P Kulkarni, [pre-print] Downregulation of KRT proteins in the Neuroendocrine System of Amyotrophic Lateral Sclerosis with PT150 as a Multi-Target Neuroprotectant, bioRxiv 2025.03.18.643927; doi: <https://doi.org/10.1101/2025.03.18.643927>

NY Flores-Perez, PS Kulkarni, M Uhart, PS Pandit, [pre-print] Shifting human-rodent interfaces under climate change: modeling the distribution of the reservoir for Junin virus and associated drivers, bioRxiv 2024.06.24.600371; doi: <https://doi.org/10.1101/2024.06.24.600371>

Kulkarni, PS and Haijema, R and Hogeveen, H. and Steeneveld, W and Mourits, M., [pre-print] Economic Impacts of Constrained Replacement Heifer Supply in Dairy Herds. Available at SSRN: <https://ssrn.com/abstract=4542707> or <http://dx.doi.org/10.2139/ssrn.4542707>

P.S. Kulkarni * , M.C.M. Mourits., M. Nielen., J. Van Den Broek, H. Hogeveen And W. Steeneveld [conference paper] DEVELOPMENT OF A LOG-NORMAL ACCELERATED FAILURE TIME MODEL FOR SURVIVAL OF DUTCH DAIRY COWS UNDER ALTERING AGRICULTURAL POLICY, SVEPM 2021, Proceedings of the Society for Veterinary Epidemiology and Preventive Medicine annual meeting, ISBN 978-0-948073-60-1 <https://bit.ly/40cZXnb>

Conferences

SVEPM 2026	London, UK
Co-instructor/ Coordinator: Preconference workshop (AI and Data-Driven Epidemiology) Instructional time: 3 hours	
Western States Modeling Symposium 2025	Online
4 th annual symposium for AK, AZ, CA, HI, ID, OR, NV and WA states Oral Presentation: Estimating transmission parameters of HPAI outbreak in Dairy cows	
EEID 2025	Notre Dame, USA
Co-instructor/ Coordinator: Pre-conference workshop (AI/ ML in Disease Ecology) Instructional time: 20 hours	
Oral Presentation: AtlasArena	Sydney, Australia
17th International Symposium of Veterinary Epidemiology and Economics Oral Presentation: AtlasArena	
EEID 2024	Stanford, USA
2024 Ecology and Evolution of Infectious Diseases (EEID) Conference Poster Presentation: Climate-driven risk modeling of zoonotic arenaviral spillover infection in South America	
UC Davis Postdoctoral Symposium 2024	Davis, USA
9th annual UC Davis Postdoctoral Research Symposium Oral Presentation: Climate-driven risk modeling of zoonotic Arenaviral spillover infection in South America	
ISESSAH 2023	Helsinki, Finland
2023 International ISESSAH “Understanding stakeholder behavior and socio-economic implications of practices and policies of animal health” Oral Presentation: Economic impact of agri-environmental policies on replacement in Dutch dairy cattle	
VEEC Study Day 2022	Wageningen, the Netherlands
Joint meeting VEE-VEEC-AESA: Fewer livestock: healthier animals, healthier environment?	

Oral presentation: Dutch dairy farmers' perspectives on culling reasons and strategies	
ISESSAH 2022	Halifax, Canada
ISESSAH is partnering with ISVEE for its 2022 conference in Halifax, Canada	
Poster presentation: Survey study: Dutch dairy farmers' perspective on culling reasons and strategies	
ISVEE 2022	Halifax, Canada
16th International Symposium of Veterinary Epidemiology and Economics	
Connecting Animals, People, and their shared environments	
Oral presentation: Impact of changes in National agricultural policy on the survival of Dutch dairy cows	
** Bursary Award Winner	
SVEPM 2021	Toulouse, France/ Online
SVEPM 2021 - Annual Conference of The Society for Veterinary Epidemiology and Preventive Medicine	
Oral Presentation: Survival analysis to study replacement strategy of Dutch dairy farmers over 10 years with varying agricultural policies	
** Bursary Award Winner	
ICBB 2015	Pune, India
2nd International Conference on Biotechnology and Bioinformatics	
Poster presentation: Genotyping of HF Crossbred cattle for Beta-Casein genes using PCR-RFLP	
** Best Poster Award Winner	

Teaching Experience

1. PHR/ SPH -266: Instructor-of-record, Spring Quarter
Weill School of Veterinary Medicine, UC Davis
2. VET-401: Facilitator for Scientific Inference discussions
Weill School of Veterinary Medicine, UC Davis
 - a. Cohort of 2028, DVM
 - b. Cohort of 2029, DVM
3. MPM-204: Applied Population Statistics III - Lecturer
Weill School of Veterinary Medicine, UC Davis
 - a. Class of 2024, student evaluation: 4.85/5 (n = 19)
 - b. Class of 2025, student evaluation 4.9/5 (n = 13)
 - c. Class of 2026, ongoing
4. MPM-200: Introduction to Information Management for Epidemiologists – Guest Lecturer
 - a. Class of 2025; Fall Quarter
5. QVE-20306: Veterinary Epidemiology and Economics - TA
Wageningen Institute of Animal Sciences, Infectious disease epidemiology group, WUR
 - a. MSc, Veterinary Epidemiology, Class of 2022

Key Skills and software

Tools and Software

R, QGIS, Python, IBM-SPSS™, SAS, Bash shell, Qualtrics™, ASREML-w, Google Colab

Key Skills

Species Distribution modelling, Spatio-temporal analysis, Regression models, survival analysis (AFT, Proportional hazards, Parametric models), Supervised/ unsupervised ML modeling, herd simulation, Bayesian Approximation optimization, agent based modeling, dynamic programming, Markov Decision process, Monte Carlo simulations

Memberships/ other activities

Sigma XI full Membership (invited) since 2025

Professional Positions

October 2018-August 2019

Biostatistics consultant/ Vet consultant • Part-time consultancy • Omega Pathology Labs, India

August 2015-August 2016

Private Veterinary Practitioner • Large Ruminants • Self-Employed

Licensed Veterinary Ambulatory service in state of Maharashtra, India

Volunteer positions

Academic Literature Reviewer: ([Web of Science Peer review](#))

Reviewer for ISESSAH 2026 conference in Thailand

Postdoctoral Scholars Association Board Nominee (2024)

Open Science Community Utrecht (2024-Present),

Book club- Bayesian Statistics and Modelling Utrecht (2023-2024),

Blog writer- Academic Transfer (2019-2024),

Veterinary Science Day 2023 Utrecht organizing committee (2023)